AI based Location Tracking in WiFi Indoor Positioning Application

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***Abstract*—As the 4th Industrial Revolution proceeds, AI(Artificial Intelligence), IoT (Internet of Things), and big data are getting great attention from people. The most efficient application of the above three technologies is navigation technology. However, indoor navigation accuracy is not very accurate due to the many obstacles and walls in the indoor environment. Therefore, in this paper, we propose indoor location tracking technology based on WiFi(Wireless-Fidelity) communication to improve user's location accuracy using mobile communication technology and location tracking technology. First, the fingerprinting technique is used to establish a fingerprinting database by measuring the RSSI values for each sample point in the offline step. Afterwards, the user's location is tracked using a weighted fuzzy matching algorithm with the measured user's RSSI value and the value of the database. Then apply the PSO algorithm to further improve the accuracy of user location tracking. According to the simulation results, it can be seen that tracking using the PSO algorithm further improves the accuracy of user location over no PSO in WiFI based indoor environment.**

Keywords—AI, Indoor Positioning, Weighted Fuzzy Matching, PSO, Fingerprinting

I. INTRODUCTION

The fourth industrial revolution is rapidly progressing around the world. Accordingly, AI(Artificial Intelligence), IoT(Internet of Things), and big data technologies are regarded as core technologies of the fourth industrial revolution. Today, most of the world's population owns smartphones, the leading IoT devices. In addition, a lot of data is stored and used importantly through big data technology. These technologies can be combined with AI technology to be used effectively in navigation.

Navigation technology basically requires the tracked current location of the user. Therefore, tracking the location of the user is very important. If you do not track the exact location of the user, the user will be taken to an inefficient route to the destination. In the current outdoor environment, GPS(Global Positioning System) technology allows the user to track the location of the user relatively accurately[1]. However, in indoor environments, GPS technology cannot be applied due to signal obstructions due to many obstacles and walls. Therefore, indoor navigation technology uses both mobile communication technology and location tracking technology. The most commonly used communication technologies include WiFi(Wireless-Fidelity), Bluetooth, and Ultra-Wide Band (UWB). Location tracking technologies include fingerprinting, triangulation and PSO(Particle Swarm Optimization).

WiFi and Bluetooth use existing RF(Radio Frequency) bands to provide services to users. At this point, the system can use the location tracking technology to track the location of the user using RSSI (Received Signal Strength Indicator) values for the location of the user. In addition, increasing the number of APs for location tracking can improve the accuracy of location tracking.

In this paper, we propose a method to track the user's location in the WiFi communication environment. WiFi is a communication method using the RF band, and APs(Access Point) are deployed in most indoor environments. However, many obstacles and walls in the indoor environment limit the ability to track a user's location with a single algorithm. Therefore, you must use a combination of location tracking technologies. So this paper uses weighted fuzzy matching algorithm and PSO algorithm at the same time.

This paper uses fingerprinting algorithm as location tracking technology in WiFi environment. Fingerprinting algorithms can be largely divided into offline sampling step and online positioning step. In the offline step, we measure the RSSI values for each sample point and then build a fingerprinting database. In the online step, the user's RSSI value is measured and the user's location is tracked through weighted fuzzy matching algorithm[2][3]. The PSO algorithm is then applied to improve the accuracy of the tracked location.

The PSO algorithm uses intelligent particles to find the best position. All particles are searched and share their best position and best position among all particles in the swarm. Based on the shared information, the particles determine their direction of movement. Therefore, every particle must periodically update the individual's optimal position and the swarm's optimal position. If the optimal position is not updated, all particles will gather in the incorrect area, causing extreme positional errors.

First, the fingerprinting algorithm performed in the offline step has the advantage that each AP only knows the RSSI value for the SP(Sample Point). And since you only need existing data to track your location, no additional devices are needed. Also, the more SPs, the better the accuracy of location tracking. However, it takes a long time to build a database. In addition, if the indoor environment changes, the database must be rebuilt. This significantly increases the cost of location tracking systems. Therefore, it is more efficient to reduce tracking position errors in the online positioning step than in the offline step[5].

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In this paper, we use weighted fuzzy matching algorithm and PSO algorithm in order for online positioning. First, the weighted fuzzy matching algorithm reduces the tracking position error by re-tracking the exact position of the user by using RSSI data in the four adjacent sample point positions obtained through the fingerprinting algorithm. Then use the PSO algorithm to improve the tracking position error once more.

This paper is organized as follows. Section II describes the system model. Section III describes the weighted fuzzy matching algorithm. Section IV describes the PSO algorithm. Section V compares the location errors before and after applying the PSO algorithm in a WiFi environment through simulation. Finally, Section VI concludes the paper.

II. System Model

In this paper, we track user location based on RSSI value in WiFi communication. RSSI can be obtained by the following equation[6].

$$RSSI_l = 10\log(P_0) - 10nlog\left(\frac{l}{l_0}\right) + N \qquad (1)$$

Where $RSSI_l(dBm)$ and $P_0(dBm)$ are the received power between the AP and the user for distance l and l_0, respectively. n is the path loss exponent and N is the noise.

The proposed environment is an empty space of 8m x 8m. In the room, four WiFi APs are placed at each vertex. The offline sample points are evenly arranged in total of 9. This can be seen in Fig. 1. In Fig. 1, the green triangle represents the AP, the purple square represents the UE to be tracked, and the red dot represents the uniformly arranged SP.

III. Weighted Fuzzy Matching Algorithm

The weighted fuzzy matching algorithm is based on a fingerprinting database. Therefore, the fingerprinting map should be constructed in the offline step. Fingerprinting maps are represented in the form of dots in the indoor environment. The points are located on the x-y coordinate plane at regular intervals. As shown in Fig. 1, the fingerprinting map is created by finding sample points located at a certain location and measuring the RSSI values between each AP and sample points using (1)[4]. As a result, the off-line database of RSSI-based location fingerprinting is expressed in H^{RSSI} as shown in the following equation and can be defined as a CIR matrix.

$$H^{RSSI} = \begin{bmatrix} h_1^1 & \cdots & h_1^m & \cdots & h_1^M \\ \vdots & & \vdots & & \vdots \\ h_n^1 & \cdots & h_n^m & \cdots & h_n^M \\ \vdots & & \vdots & & \vdots \\ h_N^1 & \cdots & h_N^m & \cdots & h_N^M \end{bmatrix} \qquad (2)$$

Here, h_n^m represents a RSSI value between the m-th AP and the n-th SP. Then calculate the vector value between each AP and SP. The calculated vector value is defined as D and can be expressed as a matrix as shown below.

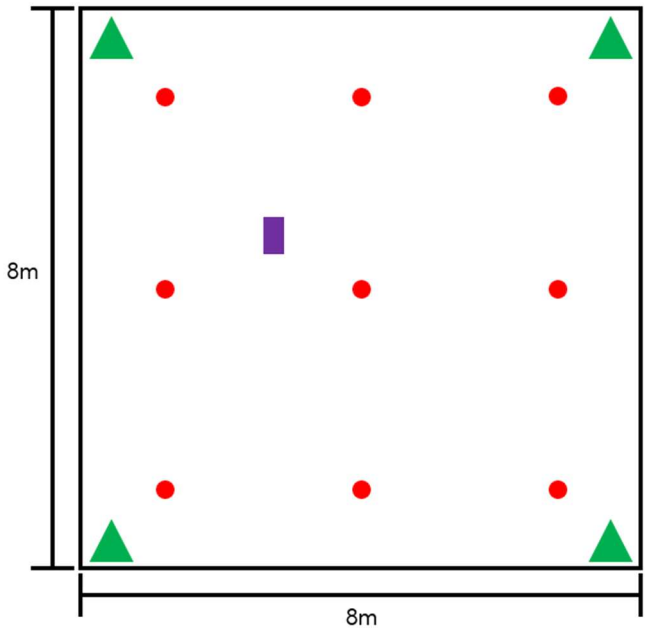

Fig. 1. System Model

$$D = \begin{bmatrix} d_1^1 & \cdots & d_1^m & \cdots & d_1^M \\ \vdots & & \vdots & & \vdots \\ d_n^1 & \cdots & d_n^m & \cdots & d_n^M \\ \vdots & & \vdots & & \vdots \\ d_N^1 & \cdots & d_N^m & \cdots & d_N^M \end{bmatrix} \qquad (3)$$

d_n^m represents a vector value between the m-th AP and the n-th SP.

The RSSI value between UE(User Equipment) k located at an arbitrary coordinate and the AP serving UE k is calculated as $Z_k^{RSSI} = [h_k^1, h_k^2, h_k^3, \ldots, h_k^M]$ to be calculated. After assessing the correlation between Z_k^{RSSI} and H^{RSSI}, you can obtain the Euclidean distance. For the j-th AP, the correlation between the fingerprint at position k in the on-line step and the fingerprint at position off is given by $\rho_{k,i}^j$, where $0 \le \rho_{k,i}^j \le 1$. Therefore, the minimum distance can be defined as

$$d_{k,i} = \|p_{k,j} - I\| = \sqrt{\sum_{j=1}^{M}\left(\rho_{k,i}^j - 1\right)^2} \qquad (4)$$

The result obtained by the above equation is the Euclidean distance vector $d_k = [d_{k,1}, d_{k,2}, \cdots d_{k,N}]$. Next, four fingerprinting vectors having high proximity with UE k as the current tracking target are selected. Proximity is calculated from high to low here. Then, the selected sample fingerprinting values may be defined as $\alpha_1, \alpha_2, \alpha_3$, and α_4 in close order, respectively. And the coordinates X_0, Y_0, Z_0 of the UE can be calculated as follows.

$$X_0 = \sum_{n=1}^{4} W_n X_n \qquad (5)$$

$$Y_0 = \sum_{n=1}^{4} W_n Y_n \qquad (6)$$

$$Z_0 = \sum_{n=1}^{4} W_n Z_n \qquad (7)$$

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W_i represents a proximity weighting factor corresponding to sample point n [3].

$$W_n = \frac{\alpha_n}{\alpha_{sum}}, \ \alpha_{sum} = \sum_{n=1}^{4} \alpha_n \tag{8}$$

IV. Application of PSO

` PSO is an intelligent evolutionary computation algorithm proposed in 1995 by James Kennedy and Russell Eberhart. PSO is a population-based probabilistic approach and is used to optimize nonlinear problems. The algorithm proceeds as follows.

Initially all particles are initialized. The particles are then randomly distributed in the search region. The distributed particles are iterated to find the optimal solution. At each iteration, the particles follow the individual's optimal position $pbest$ and the optimal position $gbest$ of all the particles in the population. Each particle produces an optimal solution based on the constantly updated $pbest$ and $gbest$. The iterative process is as follows.

$$V_i(\tau+1) = wV_i(\tau) + c\,r\,[pbest_i(\tau) - x_i(\tau)] + c\,r\,[gbest(\tau) - x_i(\tau)] \tag{9}$$

$$X_i(\tau+1) = X_i(\tau) + V_i(\tau+1) \tag{10}$$

$$w = w_{max} - \frac{\tau}{\mathrm{T}}(w_{max} - w_{min}) \tag{11}$$

Where $V_i(\tau)$ is the velocity of the i-th particle in the τ-th iteration. $X_i(\tau)$ is the position of the i-th particle in the τ th iteration. And w is the inertia coefficient, c is the acceleration coefficient, and r is any shrinkage coefficient. τ is the current number of iterations and T is the total number of iterations of the algorithm.

V. Simulation Results

The simulation was performed assuming an empty space of 8m x 8m as mentioned in the system model. Important parameters of the system are summarized in Table I. In addition, as shown in Fig. 1, a total of four WiFi APs are placed at each vertex. In addition, in order to reduce the impact of the off-line step in the user location tracking technique, a total of nine sample points are set and are evenly spaced.

TABLE I. Simulation Parameter

Parameter	Value
Room Size	8m x 8m x 3m
Number of repeated counts	10,000
Number of AP	4
Number of Sample Point	9
Number of Particle	10
Power per each AP	20W

First, each AP establishes an offline fingerprinting database by measuring RSSI values for respective SPs. Thereafter, in the online positioning step, four APs measure an RSSI value for a location of a UE indicated by a purple rectangle. As shown in Fig. 2, we set the four sample points with the highest proximity marked in yellow. Thereafter, weights of adjacent sample points may be calculated through a weighted fuzzy matching algorithm to derive estimated coordinates of the UE indicated by the green points.

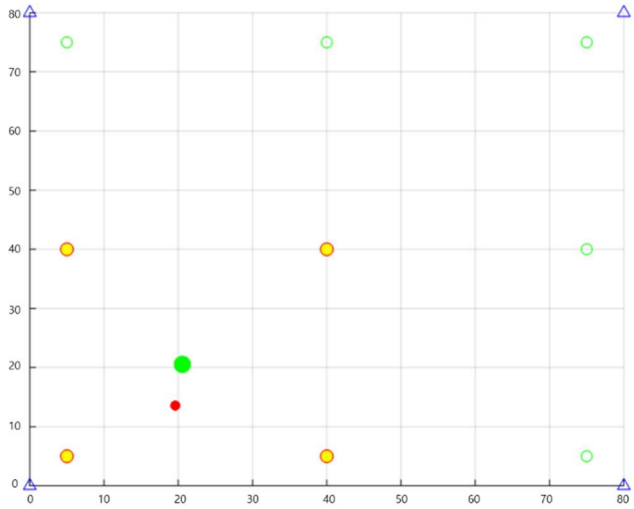

Fig. 2. Location Estimation using weighted Fuzzy Matching

Figure 3 shows the simulation results of PSO algorithm after weighted fuzzy matching algorithm. The experiment limits the region to four adjacent sample points obtained

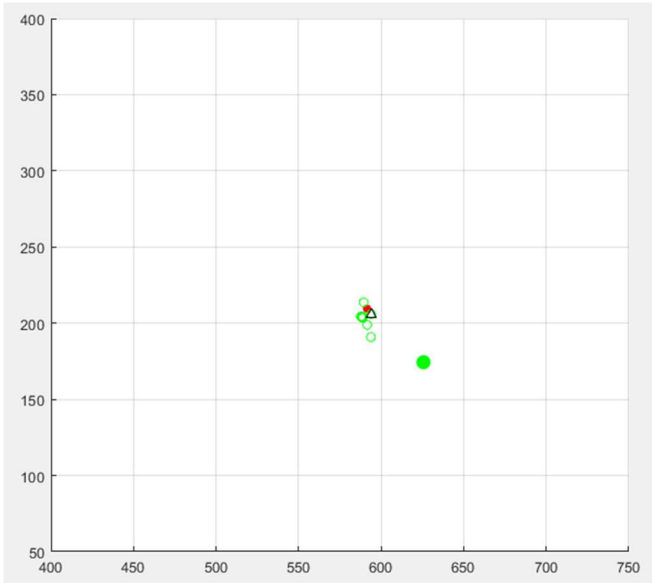

Fig. 3. Location Estimation with PSO

through weighted fuzzy matching. Therefore, the average position error is further improved by limiting the initial area of the PSO algorithm. The position error can be obtained by calculating the difference between the actual UE position and the estimated UE position.

In this paper, the location of the UE was randomized and a total of 10,000 simulations were performed. After the simulation, the average position error of the UE coordinates for a total of 10,000 times was measured. The results are shown in Table II.

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TABLE II. SIMULATION RESULTS

Parameter	Value	
	No PSO	With PSO
Average Error	2m	1.2m

Table II summarizes the average position error of the UE coordinates through a total of 10,000 simulations in the WiFi environment. The result is the result of applying the existing fingerprinting technique and the weighted fuzzy matching algorithm, and improving the position accuracy by additionally using the PSO algorithm. As shown in Table 2, when the PSO algorithm was not applied, an average position error of about 2 m occurred. However, when the PSO algorithm is applied, it is confirmed that an average position error of about 1.2m occurs. From the above results, we confirmed that the location accuracy of the PSO algorithm was higher than that of the single algorithm in the environment with few sample points.

VI. CONCLUSION

Advances in GPS technology enable relatively accurate user location tracking in outdoor environments. However, due to signal loss, there is a limit to apply in a complex indoor environment. So in indoor environments, the user's location is tracked using communication technologies such as WiFi, Bluetooth, and Ultra-Wide Band (UWB). However, due to some obstacles and walls in the indoor environment, there is a limit to tracking the exact location of the user. Therefore, when using location tracking technology, it is possible to achieve more improved location accuracy by combining several algorithms rather than using a single algorithm. This paper uses a combination of fingerprinting technique, weighted fuzzy matching algorithm and PSO algorithm. Simulation shows that the proposed method can achieve higher position accuracy.

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